



COMPONENTS

IoT chip

u-blox, a global provider of wireless communication technologies, presents the SARA-R410M-63B module for LTE-M cellular network operators. The module series supports a broad spectrum of



Internet of Things (IoT) applications such as smart metering, telematics, asset and vehicle

tracking, security systems and building automation, as well as smart lighting solutions and connected health.

The SARA-R410M-63B modules include an essential root of trust (RoT) that provides an advanced set of security functionalities, making them ideal for mission-critical IoT solutions. It also has an extended operating temperature range of -40°C to $+85^{\circ}\text{C}$ for optimal industrial applications.

Apart from the above features, u-blox cellular positioning technology enables standalone location estimation based on surrounding mobile network information or in conjunction with Global Navigation Satellite System (GNSS) positioning data to improve the location.

u-blox

www.u-blox.com

MCU with Soteria-G2

With the rapid growth of 5G cellular infrastructure, networks and data centres supporting the expansion of



cloud computing are growing. This has prompted developers to seek new ways of ensuring that

operating systems remain secure and uncompromised.

Hence, Microchip Technology has announced a new cryptography-enabled microcontroller (MCU), the CEC1712 MCU with Soteria-G2 custom firmware. It is designed to stop malicious malware such as rootkit and bootkit for systems that boot from external Serial Peripheral Interface (SPI) flash memory.

Microchip's Soteria-G2 custom firmware features CEC1712 Arm Cortex-M4 based MCU that provides the secure boot with hardware root of trust protection in a pre-boot mode. This is a great feature security-wise for operating systems that boot from external SPI flash memory. Additionally, the CEC1712 provides key and code rollback protection during operation lifetime, thereby enabling in-field security updates.

Microchip

www.microchip.com

Wireless charging IC

In the age of 5G communication with an ever-increasing demand for more power and higher efficiency, STMicroelectronics' STWLC68 product family provides the best solution for wireless-



charging applications with up to eighty per cent efficiency, high power transfer and safety.

The new product family can operate both as a high-power receiver and a transmitter for enabling rapid power transfer and power-sharing with Foreign Object Detection (FOD) and other IPs. ST's proprietary high-voltage technology, paired with excellent mixed-signal design and high-quality assurance, enables the customers to

deliver cutting-edge wireless-charging products.

The STWLC68 family of highly integrated devices needs a very low external bill of materials (BoM), ideal for integration in a wide range of applications. It's compliant with WPC Qi 1.2.4 standard and compatible with all Qi-certified devices in the market.

STMicroelectronics

www.st.com

Inductors

Vishay Intertechnology has expanded its IHLE series of low-profile, high-current inductors, which feature integrated e-shields for the reduction of electromagnetic interference (EMI)



with new commercial and automotive-grade devices. The Vishay Dale IHLE-5050FH-51 and IHLE-5050FH-5A bring down costs and save board space by eliminating the need for separate board-level Faraday shielding.

Featuring operating temperature of up to 155 degrees Celsius and coplanarity of four terminals within $\leq 100\text{ }\mu\text{m}$, the inductors are optimised for energy storage in DC/DC converters of up to 2MHz and provide excellent noise attenuation in high current filtering applications (up to the self-resonance frequency (SRF)) of the inductor.

IHLE-5050FH-51 applications include desktop PCs and servers, high current POL converters, low-profile, high-current power supplies, and battery-powered devices. The IHLE-5050FH-5A can be used in automotive engine and transmission control units, diesel injection drivers, noise suppression for motors, windshield wipers, power mirrors and seats, entertainment/navigation systems, heating and ventilation blowers, and LED and HID lighting.

Vishay Intertechnology

www.vishay.com